The Arithmetic of the QA Map: $\mathbb{Z}[\varphi]$, Norm Invariants, and Half-Pisano Orbits

[Author redacted for review]

Abstract

The Quantum Arithmetic (QA) map $T : (b, e) \mapsto (b{+}e,\ b{+}2e) \pmod{m}$ is the linear representation of multiplication by φ^2 in the ring $\mathbb{Z}[\varphi]$ of integers of $\mathbb{Q}(\sqrt{5})$, where $\varphi = (\sqrt{5}+1)/2$. The quadratic form $f(b, e) = b^2 + be - e^2$, which is the algebraic norm $N(b + e\varphi)$ in $\mathbb{Q}(\sqrt{5})$, is T-invariant; mod-m values of f classify QA orbits. Since $T = Q^2$ where Q is the Fibonacci companion matrix, the orbit length of any primitive state equals $\pi(m)/2$, half the Pisano period (Wall, 1960). The projective ratio $z_t = e_t/(b_t + e_t)$ evolves by a Möbius map with unique attracting fixed point $z^* = 1/\varphi$. Finally, the harmonic index H_{QA} used in QA-based learning admits a clean operator decomposition: $4H_{\mathrm{raw}} = \mathrm{tr}(\Lambda)$ where $\Lambda = \mathrm{diag}(\sigma_d, \sigma_{od})$ is a 2×2 cross-coupling operator; the equal-weight Rayleigh identity $2H_{\mathrm{raw}} = R(\Lambda, (1, 1)^\top/\sqrt{2})$ is canonical by a three-part symmetry argument. This note consolidates the algebraic foundations; optimization applications appear in the companion paper [2].

Contents

1 Introduction **2**

2 The Ring $\mathbb{Z}[\varphi]$ and the QA Map **2**
2.1 The golden ratio and its ring of integers . . . 2
2.2 The QA map as Q^2 . . . 2

3 The Quadratic Norm and Orbit Invariant **3**
3.1 Algebraic norm . . . 3
3.2 Geometric determinant identity . . . 3
3.3 Orbit classification for $m = 9$ . . . 3

4 Orbit Lengths and the Half-Pisano Theorem **3**
4.1 The Pisano period . . . 3

5 Projective Flow and the Golden Attractor **4**

6 The H_{QA} Operator Decomposition **5**
6.1 The cross-coupling operator . . . 5
6.2 Why the equal-weight vector $v_0 = (1, 1)^\top/\sqrt{2}$ . . . 5
6.3 Bilinear identity . . . 6

7 Conclusion **6**

1 Introduction

The Quantum Arithmetic (QA) system is a modular arithmetic framework whose dynamics arise from a simple recurrence: given a pair $(b, e) \in \{1, \ldots, m\}^2$, one step produces $(d, a) = (b + e \bmod m, b + 2e \bmod m)$, with the convention that 0 is replaced by m. The resulting sequences are Fibonacci-like modulo m, and the QA map $T : (b, e) \mapsto (d, a)$ generates a partition of the state space into finitely many closed orbits.

At first glance this is a piece of recreational combinatorics. The purpose of this note is to show that the structure is, in fact, determined entirely by the arithmetic of $\mathbb{Q}(\sqrt{5})$: the map is multiplication by φ^2, the invariant of each orbit is the algebraic norm, and orbit lengths follow from the Pisano period. These are classical facts in algebraic number theory, here made explicit for the QA setting.

The note is organised as follows. Section 2 defines the ring $\mathbb{Z}[\varphi]$ and identifies T as Q^2. Section 3 proves norm invariance and gives the geometric determinant interpretation. Section 4 recalls the Half-Pisano theorem and derives orbit lengths. Section 5 analyses the projective Fibonacci flow. Section 6 derives the H_{QA} operator decomposition. Section 7 concludes with directions for the companion paper.

2 The Ring $\mathbb{Z}[\varphi]$ and the QA Map

2.1 The golden ratio and its ring of integers

The golden ratio $\varphi = (\sqrt{5} + 1)/2$ satisfies $\varphi^2 = \varphi + 1$ and is a root of the minimal polynomial $x^2 - x - 1$. Its Galois conjugate is $\bar{\varphi} = (1 - \sqrt{5})/2 \approx -0.618$, satisfying $\varphi + \bar{\varphi} = 1$ and $\varphi\bar{\varphi} = -1$.

The ring of integers of $\mathbb{Q}(\sqrt{5})$ is $\mathbb{Z}[\varphi] = \{a + b\varphi : a, b \in \mathbb{Z}\}$, with addition and multiplication inherited from $\mathbb{R}$. Multiplication by φ acts on the coordinate pair (a, b) as:

$$\varphi(a + b\varphi) = a\varphi + b\varphi^2 = a\varphi + b(\varphi + 1) = b + (a + b)\varphi,$$

so the matrix of $\times\varphi$ in the basis $\{1, \varphi\}$ is

$$Q = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

This is the Fibonacci companion matrix: if (F_n) denotes the Fibonacci sequence then $Q^n = \left(\begin{smallmatrix} F_{n-1} & F_n \\ F_n & F_{n+1} \end{smallmatrix}\right)$.

2.2 The QA map as Q^2

Proposition 2.1 (QA map is $\times\varphi^2$)**.** *Let $T : (b, e) \mapsto (b{+}e,\ b{+}2e) \pmod m$ be the QA map (with 0 replaced by m). Then $T = Q^2$ as a map on $(\mathbb{Z}/m\mathbb{Z})^2$, and corresponds to multiplication by $\varphi^2 = \varphi + 1$ in $\mathbb{Z}[\varphi]/m\mathbb{Z}[\varphi]$.*

Proof. Direct computation:

$$Q^2 = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}^2 = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

Applied to $(b, e)^\top$: $Q^2(b, e)^\top = (b + e,\ b + 2e)^\top = (d, a)^\top$. The ring interpretation: $(b + e\varphi) \cdot \varphi^2 = (b + e\varphi)(\varphi + 1) = b\varphi + b + e\varphi^2 + e\varphi = b + e + (b + 2e)\varphi = d + a\varphi$. $\square$

Remark 2.2. The map Q^1 would send $(b, e) \mapsto (e, b + e)$, i.e., one Fibonacci step. $T = Q^2$ skips the intermediate pair and maps directly to the pair two steps ahead. The QA system thus "runs the Fibonacci recurrence at double speed".

3 The Quadratic Norm and Orbit Invariant

3.1 Algebraic norm

The algebraic norm in $\mathbb{Q}(\sqrt{5})$ is $N(\alpha) = \alpha\bar{\alpha}$ for $\alpha = a + b\varphi$:

$$N(a + b\varphi) = (a + b\varphi)(a + b\bar{\varphi}) = a^2 + ab(\varphi + \bar{\varphi}) + b^2\varphi\bar{\varphi} = a^2 + ab - b^2.$$

For a QA state (b, e) we define $f(b, e) = N(b + e\varphi) = b^2 + be - e^2$.

Proposition 3.1 (f is T-invariant)**.** *For any $(b, e) \in \mathbb{Z}^2$, $f(T(b, e)) = f(b, e)$.*

Proof. Since T is multiplication by φ^2 and the norm is multiplicative:

$$f(T(b, e)) = N\big((b + e\varphi)\varphi^2\big) = N(b + e\varphi) \cdot N\big(\varphi^2\big) = f(b, e) \cdot N(\varphi)^2 = f(b, e) \cdot (-1)^2 = f(b, e). \quad \square$$

3.2 Geometric determinant identity

The norm has a direct geometric meaning in terms of the QA step matrix.

Proposition 3.2 (Determinant identity)**.** *Let $M = \left(\begin{smallmatrix} b & e \\ d & a \end{smallmatrix}\right)$ where $(d, a) = T(b, e)$ in $\mathbb{Z}^2$ (before reduction mod m). Then* $\det(M) = f(b, e)$.

Proof. $\det(M) = ba - ed = b(b + 2e) - e(b + e) = b^2 + 2be - eb - e^2 = b^2 + be - e^2 = f(b, e)$. $\square$

3.3 Orbit classification for $m = 9$

For $m = 9$ the state space is $\{1, \ldots, 9\}^2$. The 3-adic valuation $v_3(f(b, e) \bmod 9)$ determines the orbit type:

Table 1: Orbit classification for $m = 9$ via the norm $f(b, e) \bmod 9$.

$v_3(f(b, e))$	Orbit length	Orbit type
≥ 4	1	Singularity (fixed point $(9, 9)$)
2–3	4	Satellite
0	12	Cosmos (primitive)

This classification is verified exhaustively for all 81 states of $\{1, \ldots, 9\}^2$. Primitive orbits (Cosmos) have $\gcd(f(b, e), 9) = 1$; there are 72 such states forming six 12-cycles.

4 Orbit Lengths and the Half-Pisano Theorem

4.1 The Pisano period

The *Pisano period* $\pi(m)$ is the period of the Fibonacci sequence modulo m: $F_{n+\pi(m)} \equiv F_n \pmod{m}$ for all $n \geq 0$. Equivalently, $\pi(m)$ is the order of the matrix Q in $\mathrm{GL}_2(\mathbb{Z}/m\mathbb{Z})$.

Theorem 4.1 (Half-Pisano Orbit Length, Wall [1])**.** *For $m \geq 3$, $\pi(m) \equiv 0 \pmod 2$. The order of $T = Q^2$ in $\mathrm{GL}_2(\mathbb{Z}/m\mathbb{Z})$ is $\pi(m)/2$. Consequently, the orbit length of any primitive state (b, e) under the QA map T divides $\pi(m)/2$, and equals $\pi(m)/2$ if and only if (b, e) is primitive.*

Proof. Step 1 (Pisano period is even). Wall [1] proves that for $m \geq 3$, the Fibonacci sequence mod m satisfies $F_{\pi(m)-1} \equiv 1 \pmod m$ and $F_{\pi(m)} \equiv 0 \pmod m$, which forces $Q^{\pi(m)} = I$ and, by analysis of the entry parities, $\pi(m) \equiv 0 \pmod 2$.

Step 2 (Order of Q^2). $\mathrm{ord}(Q^2) = \mathrm{ord}(Q)/\gcd(2, \mathrm{ord}(Q)) = \pi(m)/2$ (since $\pi(m)$ is even).

Step 3 (Orbit length). The orbit of (b, e) under T has length $\mathrm{ord}(Q^2)/\mathrm{stab}$, where stab is the size of the stabiliser. For primitive (b, e) the stabiliser is trivial, giving length $\pi(m)/2$. □

Corollary 4.2 (Pisano values for small m)**.**

m	$\pi(m)$	*Primitive orbit length* $\pi(m)/2$
3	*8*	*4*
5	*20*	*10*
7	*16*	*8*
9	*24*	*12*
24	*24*	*12*

5 Projective Flow and the Golden Attractor

Proposition 5.1 (Projective Fibonacci flow)**.** *Let $z_t = e_t/(b_t + e_t)$. The projective ratio evolves by the Möbius map*

$$z_{t+1} = \frac{1 + z_t}{2 + z_t}$$

with unique attracting fixed point $z^* = (\sqrt{5} - 1)/2 = 1/\varphi \approx 0.618$. The convergence rate near z^* is $|z_{t+1} - z^*| \approx |z_t - z^*|/\varphi^2$.

Proof. From $(d, a) = (b + e, b + 2e)$:

$$z_{t+1} = \frac{e_{t+1}}{d_{t+1} + e_{t+1}} = \frac{a}{d + a} = \frac{b + 2e}{(b + e) + (b + 2e)} = \frac{b + 2e}{2b + 3e}.$$

Dividing numerator and denominator by $b + e$ and setting $z = e/(b + e)$:

$$z_{t+1} = \frac{1 + z}{2 + z}.$$

Fixed points: $z^* = (1 + z^*)/(2 + z^*) \Rightarrow (z^*)^2 + z^* - 1 = 0 \Rightarrow z^* = (\sqrt{5} - 1)/2 = 1/\varphi$. Linearising about z^*: derivative $(2 + z^* - 1 - z^*)/(2 + z^*)^2 = 1/(2 + z^*)^2 = 1/\varphi^4$...

More directly: near z^*, $z_{t+1} - z^* \approx f'(z^*)(z_t - z^*)$ where $f(z) = (1 + z)/(2 + z)$ gives $f'(z) = 1/(2 + z)^2$. At $z^* = 1/\varphi$, $2 + z^* = 2 + 1/\varphi = \varphi + 1 + 1/\varphi = (\varphi^2 + 1)/\varphi$, and $f'(z^*) = \varphi^2/(\varphi^2 + 1)^2$. Since $\varphi^2 = \varphi + 1$ and $\varphi^2 + 1 = \varphi + 2$, $f'(z^*) = (\varphi + 1)/(\varphi + 2)^2 < 1$, confirming z^* is attracting. □

Remark 5.2. In the unmodded (linear) case, $\sigma_d = \cos(2 \arctan(z_t))$ converges to $\cos(2 \arctan(1/\varphi))$ as $t \to \infty$, providing a computable limiting value for the degree-0 component of H_{raw}.

6 The H_{QA} Operator Decomposition

The harmonic index H_{QA} used in QA-based optimisation is defined via the cross-coupling operator of the QA step matrix M.

6.1 The cross-coupling operator

Definition 6.1 (QA step matrix and cross-coupling)**.** Given a state (b, e) with one QA step $(d, a) = T(b, e)$ (in $\mathbb{Z}$, before reduction mod m), define the step matrix $M = \left(\begin{smallmatrix} b & e \\ d & a \end{smallmatrix}\right)$ and the two coupling ratios:

$$\sigma_d(M) = \frac{M_{11}M_{22}}{M_{12}^2 + M_{21}^2} = \frac{ba}{e^2 + d^2}, \quad \text{(degree-0, scale-invariant)}$$
$$\sigma_{od}(M) = \frac{M_{12}M_{21}}{M_{11} + M_{22}} = \frac{ed}{b + a}. \quad \text{(degree-1)}$$

Set $\Lambda = \mathrm{diag}(\sigma_d, \sigma_{od})$ and $H_{\mathrm{raw}} = \mathrm{tr}(\Lambda)/4$. The bounded harmonic index is $H_{\mathrm{QA}} = |H_{\mathrm{raw}}|/(1 + |H_{\mathrm{raw}}|) \in (0, 1)$.

Proposition 6.2 (Trace and Rayleigh identities)**.** *[(i)]*

1. $4H_{\mathrm{raw}} = \mathrm{tr}(\Lambda) = \sigma_d + \sigma_{od}$.
2. $2H_{\mathrm{raw}} = R(\Lambda, v_0)$ *where* $v_0 = (1, 1)^\top/\sqrt{2}$ *is the equal-weight unit vector and* R *denotes the Rayleigh quotient.*
3. $\sigma_{od} = e/2$ *via the Fibonacci identity* $a + b = 2d$.
4. *Both identities satisfy transpose symmetry:* $\sigma_d(M) = \sigma_d(M^\top)$ *and* $\sigma_{od}(M) = \sigma_{od}(M^\top)$.

Proof. (i) Immediate from $\mathrm{tr}(\Lambda) = \sigma_d + \sigma_{od}$. (ii) $R(\Lambda, v_0) = v_0^\top \Lambda v_0 = \frac{1}{2}(\sigma_d + \sigma_{od}) = 2H_{\mathrm{raw}}$. (iii) $a + b = (b + 2e) + (b) = 2(b + e) = 2d$, so $ed/(b + a) = ed/(2d) = e/2$. (iv) Swapping rows and columns of M sends $(b, e, d, a) \mapsto (b, d, e, a)$, and $ba/(d^2 + e^2) = ba/(e^2 + d^2) = \sigma_d$ (unchanged), $de/(b + a) = \sigma_{od}$ (unchanged). □

6.2 Why the equal-weight vector $v_0 = (1, 1)^\top/\sqrt{2}$

Proposition 6.3 (Canonical Rayleigh vector)**.** *The vector* $v_0 = (1, 1)^\top/\sqrt{2}$ *is the unique unit vector in the positive orthant satisfying all three of:*

[(i)]

1. ***Swap symmetry.*** v_0 *is invariant under the coordinate permutation* $(v_1, v_2) \mapsto (v_2, v_1)$*, which corresponds to interchanging the two QA pairs* $(b, e) \leftrightarrow (d, a)$*. Since* T *is a bijection on every orbit, neither pair is algebraically privileged.*
2. ***Transpose consistency.*** Λ *is symmetric under* $M \mapsto M^\top$ *(Proposition 6.2(iv)), so the natural Rayleigh vector must also be symmetric.*
3. ***Trace recovery.*** $\mathrm{tr}(\Lambda) = 2\,R(\Lambda, v_0)$*: the Rayleigh quotient at* v_0 *recovers the normalised trace of* Λ *with the unique scalar factor* $1/\sqrt{2}$ *(up to sign).*

Proof. Any unit vector $v = (\cos\theta, \sin\theta)^\top$ with $\theta \in (0, \pi/2)$ satisfies $R(\Lambda, v) = \sigma_d \cos^2\theta + \sigma_{od} \sin^2\theta$. Condition (i) requires $\cos\theta = \sin\theta$, hence $\theta = \pi/4$ and $v = v_0$. Conditions (ii) and (iii) are then verified directly. □

6.3 Bilinear identity

The following polynomial identity relates the two coupling ratios to the orbit invariant $f(b, e)$ and holds in $\mathbb{Z}$ (before reduction mod m):

$$4H_{\text{raw}} \cdot (e^2 + d^2) \cdot (b + a) = ba(b + a) + ed(e^2 + d^2). \tag{1}$$

This is an algebraic identity in the four QA values (b, e, d, a), verified directly by substituting $d = b + e$, $a = b + 2e$. It encodes the balance between the scale-free (σ_d) and amplitude (σ_{od}) components of H_{raw}.

7 Conclusion

We have shown that the QA map $T : (b, e) \mapsto (b + e, b + 2e) \pmod{m}$ is, in essence, the arithmetic of $\mathbb{Z}[\varphi]$:

- $T = Q^2$ is multiplication by φ^2 (Proposition 2.1).
- The quadratic norm $f(b, e) = b^2 + be - e^2 = N(b + e\varphi)$ is T-invariant and classifies orbits (Propositions 3.1, 3.2; Table 1).
- Primitive orbit lengths equal $\pi(m)/2$, half the Pisano period (Theorem 4.1).
- The projective ratio $z_t = e_t/(b_t + e_t)$ is attracted to $1/\varphi$ by a Möbius map (Proposition 5.1).
- The harmonic index H_{QA} has a canonical operator form $4H_{\text{raw}} = \text{tr}(\Lambda)$, and the equal-weight Rayleigh identity $2H_{\text{raw}} = R(\Lambda, v_0)$ is forced by symmetry (Propositions 6.2, 6.3).

These properties are not incidental. They form a coherent algebraic picture: the QA system is a mod-m slice of the arithmetic of $\mathbb{Z}[\varphi]$, and the harmonic index H_{QA} is the natural cross-coupling scalar of this arithmetic. Applications to convergence certificates and learning-rate optimisation are developed in the companion paper [2].

References

[1] D. D. Wall. Fibonacci primitive roots. *The American Mathematical Monthly*, 67(6):525–532, 1960.

[2] [Author redacted for review]. Unified curvature certificates across learning, aggregation, attention, arithmetic, and symbolic search. *arXiv preprint*, 2026.

[3] G. H. Hardy and E. M. Wright. *An Introduction to the Theory of Numbers*, 6th ed. Oxford University Press, 2008.

[4] N. N. Vorobyov. *The Fibonacci Numbers.* D. C. Heath, 1963.